

COMP3069 Computer Graphics

Coursework (Assessment 3) - 2024

Description

This assignment is compulsory and is worth **70%** of your final mark for COMP3069. It is due for submission by **11.59pm** of **20 December 2024 (Friday)**. Late submissions will receive a penalty of 5% of the assignment grade per day.

You should **submit two files**:

1. A zip file containing your visual studio solution, including code, textures, shader files, executable, 'include' and 'lib' directories. **50%**
2. A single pdf file containing your report (description and demonstration). **20%**

Requirements

In this coursework, you are required to apply what you have learnt about 3D computer graphics and graphics programming, to create and display a scene consisting of objects. The coursework will test your understanding of the concepts of computer graphics and your ability to implement a 3D scene. Whenever possible, please **first consider and apply** the topics, concepts and mechanisms covered in the lectures and lab sessions, including, but not limited to:

1. Graphics APIs (e.g., OpenGL, GLFW, GLSL, GLM),
2. 3D Modelling,
3. 3D Transformations,
4. Cameras,
5. Textures,
6. Lights, and
7. Anti-aliasing.

You are also required to **write a report** using the **Report Sheets** provided to outline how you have met the requirements. The report should contain screenshots of your program (coding) and output (displays) as your demonstration (10%), and description (10%).

Technical Requirements

Figure 1 shows a picture (copied from <https://www.freepik.com/free-photos-vectors/3d-object#uuid=57ea2d72-c6d6-4669-a849-453a9ddec771>) of a scene, where you can find various objects (including two balls, a ring, and a cone).



Figure 1: A scene containing 3D objects of two balls, a ring and a cone.

You are expected to use your imagination and try to be creative when designing your scene. The **technical requirements** are as follows.

1. You should create all 3D objects (including 3D modelling & 3D Transformations) and display them in your scene by referring to Figure 1. The connections and the looks of the objects should be as similar to those in Figure 1 as possible.
2. The scene should contain all objects. All objects should be animated.
3. You need to create at least three lights of different types (i.e., directional, positional and spot lights). You should be able to switch between the lights.

4. You need to find the textures that look real and are as close to those in Figure 1 as possible to colour the objects in your scene.
5. You need to have two different types of interactive cameras (i.e., model-viewer and fly-through cameras) to demonstrate your scene from different viewpoints or view directions. You should be able to switch between a model-viewer camera and a fly-through camera. You should allow user interactive viewing (using keyboard/mouse input).
6. You need to have all objects to be interactive (keyboard/mouse input) for the changes on response to the user input.
7. You need to apply anti-aliasing mechanisms to smooth the edges.

There is some **flexibility** to account for what you want to model. You might want to

1. model something by hand, or
2. procedurally generate some vertex data.

However, the more use of the knowledge that you directly learnt in COMP3069, the higher marks will be given.

Submission

Before **submitting your code**, test it using Visual Studio 2022. Marks will be lost for programs that do not compile or have issues with linking to resources.

Please submit your work by **11.59pm of 20 December 2024**. Please note that you can only submit your work once, so, after you submit your work, you cannot change and submit it again.

You are not able to submit your **self-contained solution** to Moodle because of the limit set for the file size, so please **zip your coding work including the whole set of solution**, save it at OneDrive (using your UNNC account), and then **share** it with **Yue** at Yue.Li3@nottingham.edu.cn, **Sean** at Sean.He@nottingham.edu.cn, and **Wooi Ping** at Wooi-Ping.Cheah@nottingham.edu.cn. Please also create a share link, and then **copy the link** to the text box at the submission site at Moodle.

You **must also submit your report** (with demo) in **PDF** format as a file attachment to the submission site. You may also zip your PDF file and then submit it if the file size is larger than 10MB.

You must be sure that the markers (Yue, Sean and Wooi Ping) will be able to download your zip file that you share at OneDrive using the link that you provide. You must also be sure that your markers can unzip it to a directory, open the .sln file, compile the project and run it without any further interventions.

The report is compulsory. Failure to submit a report may result in a zero mark for the whole coursework.